THE UNITED REPUBLIC OF TANZANIA

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

KIBAHA PRIVATE SECONDARY SCHOOL ASSOCIATION (KIPSSA)

FORM FOUR PRE-KIPSSA EXAMINATION

032 CHEMISTRY 1

Time: 3 Hours **2025**

Instructions

1. This paper consists of sections A, B, and C with **eleven (11)** questions.
2. Answer all questions in section A and B and **two (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, Section B carries **fifty-four (54)** marks and Section C carries **thirty (30)** marks
4. Cellular phones and any unauthorized materials are not allowed in the examination room.
5. Write your **examination number** on every page of your answer sheet(s)
6. The following constants may be used:

   Atomic mass: H=1, C=12, N=14, O=16, Na=23, Ca = 40, S=32, Mg=24, Cl=35.5, K =, Pb = 207, Fe = 56

   Avogadro's number = $6.02 \times 10^{23,}$ GMV at S.T.P = $22.4dm^3$, 1Faraday = 96,500coulombs.

   Standard pressure = 760mm Hg, Standard temperature = 273K, 1 litre = $1dm^3 = 1000cm^3$.

SECTION A (16 MARKS)

1. For each of the items (i) – (x), choose the most correct answer from the given alternatives and write its letter besides the item number in the answer sheet provided

(i) Which of the following statements is correct about an atom?

A. The protons orbit outside the nucleus in shells

B. The neutrons orbit outside the nucleus in shells

C. Electrons and neutrons are in the nucleus

D. Protons and neutrons are in the nucleus

E. Protons and electrons are in the nucleus

(ii) A Chemistry student at Seypalm Secondary School made the following statements:

I. Organic chemistry is the Chemistry of carbon compounds.

II. All living things contain carbon compounds

III. Monomers are large molecules made up of polymers

IV. Polymers are large molecules obtained from smaller organic molecules

V. Carbon is the most unusual atom

Which of the following is true about the student's statements?

A. Statement (i) is not correct

B. Statement (ii) is not correct

C. Statement (iii) is not correct

D. Statement (iv) is not correct

E. All of the student's statements are correct

(iii) An element Y is found in period 4 and group II of the periodic table. What is the combining power of the element?

A. 4 B. 2 C. 6 D. 1 E. 5

(iv) A complete fertilizer contains

A. Two micronutrients

B. Only one primary macronutrient.

C. All three primary macronutrients.

D. Two primary macronutrients

E. Two micronutrients.

(v) What is the amount of charge when a current of 2 Amperes flows through a certain point in an electrolyte for 0.5 seconds?

A. 2 coulombs

B. 2.5 coulombs

C. 10 coulombs

D. 1 coulomb

E. 1.5 coulombs.

(vi) Substance Q is a colourless liquid at room temperature, which turns white anhydrous copper (II) sulphate blue, neutral to litmus and freezes at 0 ^{o}C. Which of the following can be substance Q?

A. Alcohol

B. Ether

C. Water

D. Hydrogen

E. Ethane

(vii) The sign on a sulphuric acid bottle said it contained 49 grams of sulphuric acid. That equals to how many moles of the sulphuric acid?

A. 0.5

B. 49

C. 1

D. 0.75

E. 2

(viii) Direct exposure to UV radiations can result in to the following, except:

A. Cholera outbreak

B. Skin cancer

C. Eye damage

D. Skin rash outbreak

E. Impairment of the immune system

(ix) A property that enables iron to be used in building bridges.

A. Ductility

B. Sonority

C. Tensile strength

D. Malleability

E. Lustrous

(x) Contains iron mixed with chromium, nickel and manganese.

A. Stainless steel

B. Bronze

C. Brass

D. Silica gel

E. Galvanizing

2. Match the items in LIST A with the corresponding items in LIST B by writing the letter of the correct response besides the item number in the answer sheet provided

List A	**List B**
(i) Macromolecules made of many smaller repeating organic molecules.	A. Alkanes
(ii) The most unusual element, whose atom has the ability to join up to itself and form very long chains of atoms.	B. Carbon
(iii) Crude oils and diesel fuels	C. Oxygen
(iv) The fractions consist of mainly methane, ethane, propane, and butane.	D. alkenes
(v) Branches of the main carbon chain, responsible for the unique chemical reactions of a molecule	E. Monomers
(vi) Organic compounds with the same molecular formula but different structural formulae	F. Polymers
	G. Isomers
	H. Natural gas
	I. Functional groups
	J. Petrol

SECTION B (54 Marks)

3.(a) Identify the substance in the following information:

(i) A solid, which is yellow when hot and white when cold.

(ii) A compound which decomposes to form a black solid, brown gas, and gas that relights a glowing splint.

(iii) Solid carbonates which do not decompose on heating but are highly soluble in water.

(iv) An aqueous solution of a greenish crystalline sulphate forms a pale-green precipitate with a sodium hydroxide solution which turns brown on standing and when exposed to air.

(b) Give reasons, explain what will happen when;

(i) Barium chloride solution is added to an aqueous solution of potassium sulphate.

(ii) Silver nitrate solution is added to an aqueous solution of sodium chloride.

(iii) Conc. sulphuric acid is exposed to the atmosphere.

(iv) Iron rod exposed to the atmosphere for one month.

(v) Hydrogen gas container is left open.

4. (a) Explain the permanent hardness of water with respect to how the condition develops.

(b) The following is an illustration of a method for removing hardness of water, which is known as an ion exchange method:



Give more information that corroborates the above illustration in describing the illustrated process of removing hardness of water.

5.(a) Mixtures can be categorized as homogenous or heterogeneous; with at least one example, explain each of the two categories of mixtures.

(b) When it comes to the separation of mixtures, an appropriate method may need to be used in order to make the mixture separation process successful. One of the methods is chromatography; explain any three of its practical application in daily life.

6. (a) Differentiate between isomers and isomerism and explain whether isomers have different physical and chemical properties.

(b) Alkynes show branching isomerism and positional isomerism. Write the structural formula of three isomers of an alkyne with the molecular formula, C_5H_8.

7. (a) The solution obtained after mixing marble chips with sulphuric acid was allowed to evaporate to dryness in the same container. After the evaporation process, the container together with its contents had a total weight of 97.3g. On the next day it was found to have 98.6g.

(i) Why did the mass increase?

(ii) What is the name of the property shown by the substance left in the container?

(iii)What would happen to the rate of reaction if calcium carbonate powder was used instead of marble chips? Explain.

(b) Solution A and B were mixed in one beaker. After a few minutes, the beaker was very hot. As a Chemistry student:

(i) Give the type of the reaction and explain your answer.

(ii) How would you adjust the temperature in such a reaction to favor the forward reaction.

8. A solution is made by diluting 25 cm^3 of 0.4 M sulphuric acid to make 1000 cm^3 solution. 25 cm^3 of the solution was required to neutralize 25 cm^3 of sodium hydroxide solution. Calculate

(a) Number of moles of acid that reacted.

(b) Number of moles of base that reacted.

(c) Molarity of base.

(d) Concentration of base.

SECTION C (30 MARKS)

9. Describe hydrogen chloride gas by the following guidelines:

   (a) The raw materials for production of the gas.

   (b) Four physical properties of hydrogen chloride gas.

   (c) Draw the diagram for the preparation of hydrogen chloride gas.

   (d) Give five uses of hydrogen chloride gas.

10. Water is the most abundant liquid in nature. It occurs on land as seas, oceans, rivers and lakes, underground water and springs. However, it is dynamic in nature as exhibited by the water cycle. With diagrammatic illustrations, describe the water cycle.

11. It is undeniable fact the combustion is essential for our survival. Justify this observation by elaborating the real-life application of combustion.